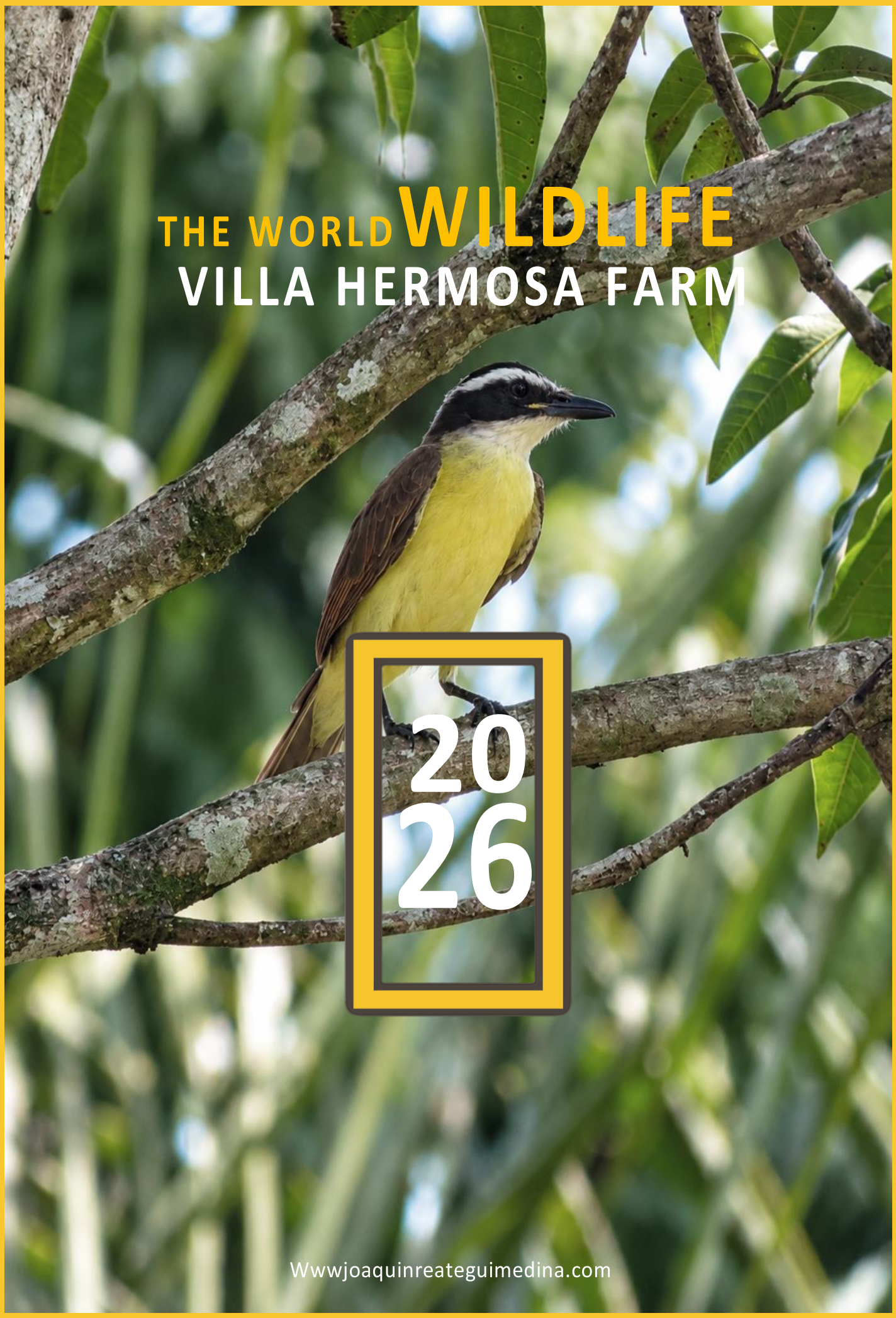

THE WORLD WILDLIFE
VILLA HERMOSA FARM
2026
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PRESENTATION

Peru possesses great cultural, historical, and natural wealth thanks to the diversity of its ecosystems, which are found in the Andes, along the coast, and throughout the Amazon rainforest. From an environmental perspective, tourism can contribute to the sustainable management of Protected Natural Areas (PNAs), raise environmental awareness, and promote the conservation of natural resources. It can also encourage the creation of new protected areas by the State.

The Loreto region offers important tourism resources and locations with great potential for developing nature tourism or ecotourism. These include the Pacaya Samiria National Reserve and the Allpahuayo Mishana National Reserve, both of which are home to extraordinary biodiversity and landscapes representative of the Peruvian Amazon.

However, many tourism resources in the rainforest are still in the development stage and face shortcomings in infrastructure, facilities, and tourism equipment. Therefore, it is essential to promote responsible planning that improves tourism services without compromising ecosystem conservation.

Flora and fauna observation is an activity focused on appreciating and studying biodiversity in its natural habitat. As part of ecotourism, it has great potential to strengthen environmental education, encourage appreciation of biological diversity, and contribute to the preservation of Protected Natural Areas.

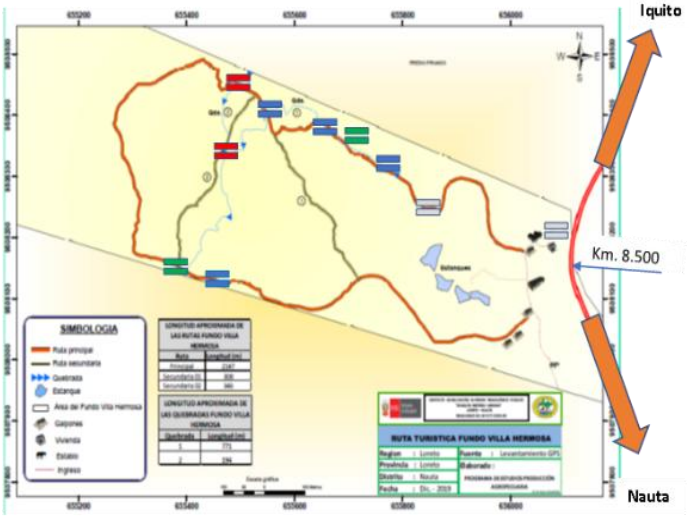
In this context, ecotourism represents an opportunity to promote the sustainable development of Loreto, generate benefits for local communities.


BIENVENIDOS AL CIRCUITO
DE INTERPRETACIÓN
ECOTURISTICA VILLA HERMOSA

VILLA HERMOSA FARM

"Protect wildlife, protect life. Every creature plays a vital role in the balance of nature."

At the research site, an ecological interpretation trail was implemented for educational purposes. This initiative was developed as part of the Official Tourism Guide program during 2013 and 2014, with the aim of promoting the creation, innovation, and development of good practices related to sustainable tourism and respect for natural resources.



Learning becomes meaningful when it enables students to solve situations they encounter in their daily lives. These learning experiences become more relevant and significant when they are designed according to students' needs and involve direct interaction with their environment. In response to this need, the Good Practices in Experiential Learning with Nature initiative emerged as an active and effective learning methodology in which the natural environment becomes a teaching resource. Its purpose is to provide students with comprehensive knowledge of ecological

interpretation, strengthen skills relevant to their professional careers, and facilitate their entry into the labor market. This methodology is based on project development, which facilitates the organization and acquisition of competencies, skills, and knowledge through real-life learning situations. Activities related to the students' context are

carried out, including direct observation of their natural environment.

Work teams consisting of administrators, teachers, and students were also organized. These teams supported the innovation of the ecological interpretation center, contributing to the achievement of the established objectives and goals.

The initial strategies adopted at the beginning of 2013 included the development of a work plan. One of the proposed projects was the creation of an ecological interpretation center at Fundo Villa Hermosa, a property belonging to the institution and located at kilometer 8.5 of the Nauta–Iquitos highway.

The project received the support of the institution's administrators because it offered an alternative learning environment where students could acquire knowledge in a natural setting and improve their guiding techniques, observation skills, and ability to explain on-site the characteristics, benefits, and proper use of local flora and fauna.

As a result, the initiative achieved first place under the title "Good Practice in Experiential Learning with Nature," in the Technical-Pedagogical Management category.

The responsible teacher was Felix Castillo Ichuta, representing the Higher Education Technological Institute (IEST) "Joaquín Reátegui Medina," in the Loreto Region.

The initiative was recognized under the distinction "Maestro que todo lo hace, Tukuy Ruraq Amauta 2014" (Master Who Does Everything).

Today, we continue working on other innovative ideas for the site in order to further implement and strengthen the knowledge and skills of future Official Tourism Guides. It is located in Peru, in the Loreto Region, Loreto Province, Nauta District.







LOCATION

Location of the Research Area

The institute owns a farm located at kilometer 8.500 of the Iquitos–Nauta Highway, covering an area of 117 hectares and 7,367 m². It is used for agricultural and livestock production, institute projects, and other activities. It also has an auditorium for educational workshops, talks, and student activities.

The “Joaquín Reátegui Medina” Public Technological Higher Education Institute, located in Nauta, Loreto, owns a farm called Villa Hermosa, covering an area of 42 hectares plus 78 m² (titled property). It is located on the right-hand side of the Iquitos–Nauta Highway, at kilometer 85.500. The area's flora and fauna can be recorded and identified to support ecotourism activities and promote environmental conservation and sustainability. The property measures 2,000 meters in depth and 200 meters in frontage. It is located in the district of Nauta, Loreto Province, Loreto Region.

Political Location

- Region: Loreto

- Province: Loreto

- District: Nauta

-Location: Villa Hermosa Farm, at kilometer 8.500 of the Nauta–Iquitos Highway.

Climate

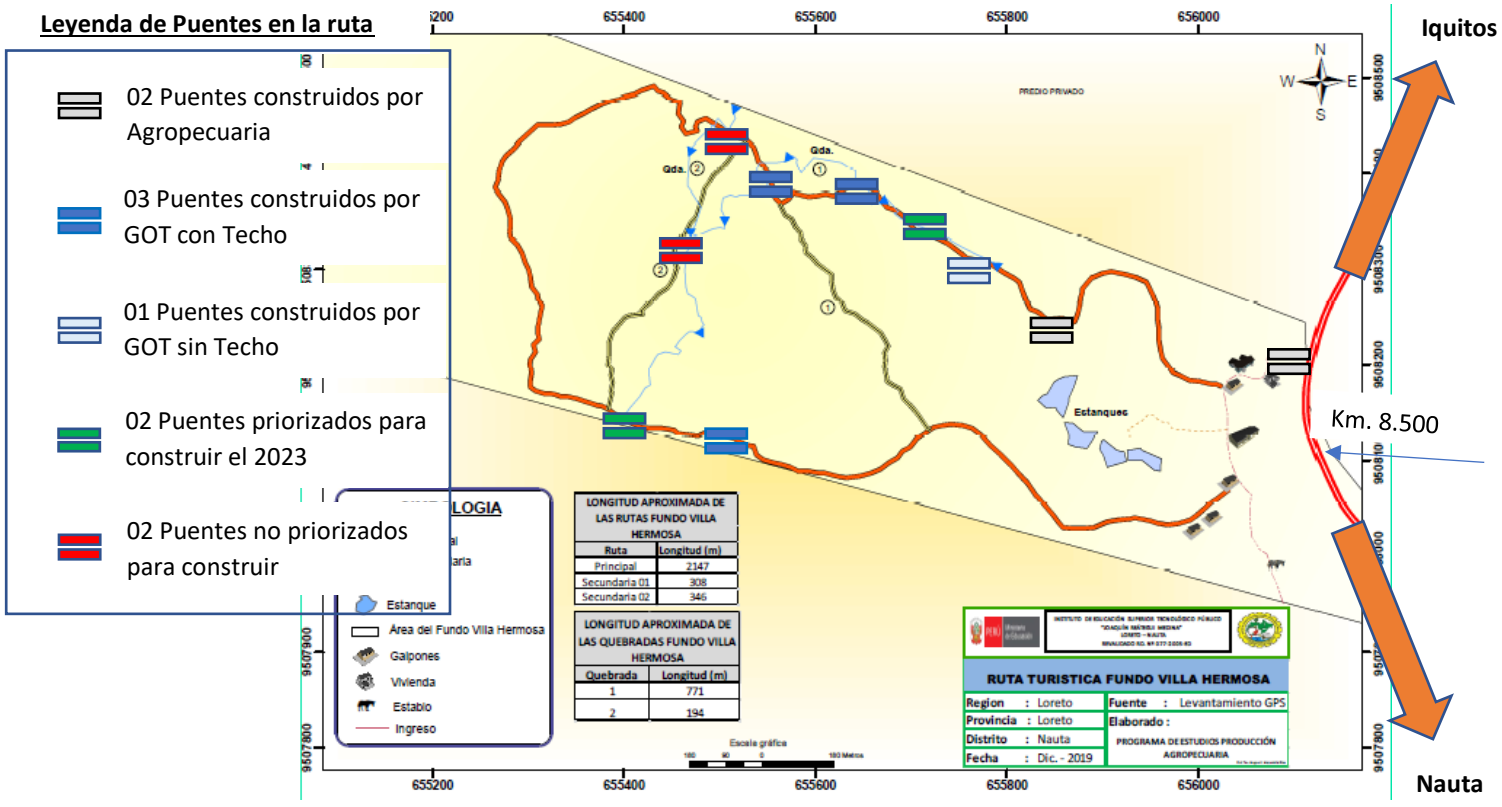
The climate is warm and rainy throughout the year. The average annual maximum temperature is 31°C (88°F), while the average annual minimum temperature is 21°C (70°F).

Topography

The study area consists of primary and secondary rainforest, with low hills and slightly undulating terrain. Along the route, there are small bodies of water known as streams.

Life Zones

The area is covered by dense vegetation and includes wetlands, tall forest, primary forest, secondary forest, and other types of natural ecosystems.





PROLOGUE

Natural

Flora: The Villa Hermosa Farm has a diversity of native plant species characteristic of the region. It can offer interpretive trails showcasing the vegetation and providing information about its importance and conservation.

Fauna: Various animal species inhabit Villa Hermosa Farm, including birds, mammals, reptiles, amphibians, and others. The center can organize birdwatching activities, nighttime tours to observe wildlife, and educational talks on the importance of conserving local fauna.

Hydrography: The presence of bodies of water, such as rivers, streams, or lagoons, can provide opportunities for environmental education activities and highlight the importance of conserving water resources.



Cultural

The Interpretation Center can provide information and experiences related to local culture, including the region's history, traditions, customs, and handicrafts. This will allow visitors to learn about and appreciate the area's rich cultural heritage.

Contemporary Technical, Scientific, and Artistic Achievements

The center can exhibit and promote projects, research, and artistic creations related to the environment and sustainability. These may include photography exhibitions, visual art, interactive installations, and other activities that highlight the importance of conservation and the relationship between human beings and nature.

Folklore

Local folklore can be presented through cultural activities such as music, traditional dances, theatrical performances, handicraft demonstrations, and local cuisine. This will provide visitors with an authentic experience and a connection to the region's cultural identity.



FLORA OF VILLA HERMOSA

The Great Biodiversity of Fundo Villa Hermosa

Fundo Villa Hermosa is a natural area located along the Iquitos–Nauta highway, in the Loreto region of the Peruvian Amazon. It covers approximately 117 hectares and is used for agricultural production, educational activities, and research projects.

One of its most important characteristics is its great biodiversity, which includes a wide variety of plants, mammals, snakes, frogs, birds, and other wildlife typical of the Amazon rainforest.

FLORA: THE DIVERSITY OF FOREST

The vegetation of Fundo Villa Hermosa represents the richness of the Amazon rainforest. It may include native trees, medicinal plants, shrubs, palms, ferns, orchids, and other tropical species. These plants provide food and shelter for animals, produce oxygen, and help maintain soil quality and ecological balance.





PRIMARY FOREST

A primary forest can vary, but the central notion is that it is a forest that has been practically unaltered by human activity for a long period of time (with the exception of use by indigenous and native communities who lead a traditional lifestyle). The FAO defines primary forests as forests of native species that regenerate naturally, where there is no clear evidence of human activities and ecological processes have not been significantly altered. The terms "primary forest" and "old-growth forest" are sometimes used interchangeably, although there is no broad consensus on the definition of the latter, and many forest scientists consider that it is possible for forests that have been altered to retain old structural and functional characteristics.Brazil is the country with the most primary forests (about 477 million hectares), followed by the Russian Federation (256 million hectares) and Canada (165 million hectares).Primary Forests are large extensions of forest ecosystems that are witnesses to what the original forests of the planet were like and that have not been transformed or altered by human activity. They have vital importance for the planet and its inhabitants, but they are disappearing rapidly and irreversibly: 80% of these forests have already been destroyed or altered. The remaining 20% is threatened by various causes (oil exploitation, mining, large dams, infrastructure), with large-scale logging, much of it illegal, being the main cause of threat at a global level. Currently, primary forests cover only 7% of the Earth's surface.

PRIMARY FOREST

Recent data reveals a previously unnoticed phenomenon: alongside the transformation of primary forests to other land uses, farmers and ranchers have allowed the reversion of important and growing areas to secondary forests. Secondary forests possess biophysical characteristics in harmony with forest management, such as high productivity and an ecologically uniform composition of dominant tree species, which simplifies their use and facilitates their silviculture, in addition to their high value in non-timber products and biodiversity. Evidence shows us how secondary forests, originated by human intervention, can be managed to provide many ecological and economic services originally provided by primary forests.We find various definitions of secondary forests in the humid tropics. The common feature to any of them is the disturbance of the ecosystem, caused or originated naturally by atmospheric, geological, wildlife phenomena, among others, or by man. In this case we talk about disturbances of anthropic origin. These are, by far, more common and occupy today a larger area than natural disturbances. Furthermore, they have important implications for land use, rural development and conservation of natural resources in general. Secondary forest is defined as woody vegetation of successional character that develops on land originally destroyed by human activities. Its degree of recovery will depend largely on the duration and intensity of previous use for agricultural crops or pastures, as

well as on the proximity to seed sources to recolonize the altered area. It is worth clarifying the basic distinction between successional secondary forests and residual forests. Residual forests are essentially primary - tall, mature or dense forests - that still retain the structure and floristic composition of an unlogged primary forest, since timber extraction - as the main product - has not drastically modified them. This distinction is obvious under the prevailing conditions in the Neotropics, where timber extraction is still highly selective, unlike what happens in many forests of Southeast Asia, where the intensity of harvesting is several times greater and, theref





MAURITIA FLEXUOSA F. (ARECACEAE)

MORICHE PALM TREE

SCIENTIFIC NAME:Mauritia flexuosa

ENGLISH NAME: Moriche Palm tree/ Buriti Palm tree

COMMON / LOCAL NAME: Aguaje

The moriche palm, Mauritia flexuosa, is a palm tree that can reach up to 35 meters in height, with a trunk diameter of about 60 centimeters. It is locally known as "aguaje". The palm produces an oval, reddish-brown fruit.

There are several local varieties of aguaje, including:

* Shambo

* Amarillo or Ponguete

* Rojizo

The aguaje fruit is a very important economic resource in the Loreto region. It is highly marketable and widely consumed, and plays a key role in the local economy.

The fruit is usually eaten fresh and is used to make a variety of desserts and drinks. For example, aguajina is a traditional Peruvian drink made with aguaje pulp diluted in water with sugar. The fruit is also used to make ice cream. In Loreto, a traditional ice cream called curichi is very popular.

In the cosmetics industry, aguaje oil extracted from the fruit pulp is used to make products such as lipstick and shampoo.

Aguaje is one of the most nutritious tropical fruits known to science. It contains high levels of vitamins, lipids, proteins, and minerals. Its edible oil has one of the highest concentrations of provitamin A found in nature.

Another food product associated with the moriche palm is the larvae of the beetle which grows inside the rotting trunk, known as suri (Rhynchophorus YUpalmarum). They live in rotting palm trunks and are considered a local delicacy. Some people compare the taste to salted butter.





BACTRIS GASIPAES

PIJUAYO PALM TREE

SCIENTIFIC NAME: Bactris gasipaes

COMMON NAME: Pijuayo

NAME IN ENGLISH: Peach Palm Tree / Pijuayo Palm Tree

SPANISH: Pijuayo

ENGLISH: Peach

FOODS:

* Fruit: Peach palm fruit / Pijuayo

* Palm Heart: Chonta

The pijuayo palm can grow up to 20 meters tall, approximately 66 feet, with a trunk diameter of 15 to 30 centimeters. Its scientific name is Bactris gasipaes. It is characterized by a trunk covered in spines up to 5 centimeters long.

The pijuayo produces an edible, oval-shaped fruit with yellow to reddish-orange skin. It has the same name as the tree, "pijuayo," and is usually consumed boiled. The fruit is also used to make drinks, such as masato de pijuayo, among others.

The trunk of this palm is used to build walls and floors in some rural communities. However, it is not as resistant as other palm species. The young leaves are used to make handicrafts such as:

* Fans

* Wallets

* Hats

It is an important economic resource for Indigenous communities, whose main activity is the sale of handicrafts.

Historically, its spines were used to make hunting weapons such as spears, arrows, and blowguns.

Inside the trunk is another important food source: the chonta, or palm heart. It is edible and is often called "jungle spaghetti." It is commonly used in regional celebrations to prepare traditional dishes such as:

* Juane de chonta

* Salads, and more.





ANACARDIUM OCCIDENTALE

THE CASHEW FRUIT

SCIENTIFIC NAME: Anacardium occidentale

NAME IN ENGLISH: Cash Nut / Cashew Apple

COMMON / LOCAL NAME: Marañón / Casho

SPANISH: Marañón / Casho

EDIBLE PARTS:

* Cashew Nut

* Cashew Apple / Fruit

TRADITIONAL MEDICINAL USES:

* Bark

* Leaves

* Nut

The marañón tree, commonly known as "casho" in the Amazon, is scientifically named Anacardium occidentale. In English it is called cashew.

The cashew tree is relatively small, reaching 3 to 5 meters in height. It produces a delicious, edible fruit called the cashew apple. The fruit is eaten when ripe, and it is also used to make a very popular non-alcoholic drink. Traditionally, the fruit is also used to prepare a tonic.

The leaves and bark are used in traditional medicine to treat conditions such as:

* Irritations

* Inflammations

* Diabetes





ASTROCARIUM CHAMBIRA

THE CHAMBIRA PALM TREE



SCIENTIFIC NAME:Astrocaryum chambira

NAME IN ENGLISH: Chambira Palm / Buriti Palm

COMMON / LOCAL NAME: Chambira

SPANISH: Chambira

FOOD:

* Chambira Fruit

TRADITIONAL MEDICINAL USES:

* Tender Fruit / Endosperm

* Root

The chambira palm grows 20 to 30 meters tall and has a trunk diameter of 20 to 35 centimeters. Its scientific name is Astrocaryum chambira. This palm is characterized by its spiny trunk, which is covered in long spines measuring 15 to 25 centimeters.

The leaves of the chambira palm are used to make handicrafts such as:

* Fans

* Wallets / shicras bags

* Hammocks

* Hats

The tree also produces a fruit similar to a small coconut. When the fruit is tender, its endosperm is used in traditional medicine to help treat diabetes.



CECROPIA SPP

THE CETICO TREE

SCIENTIFIC NAME:Cecropia

ENGLISH NAME: Cecropia Tree / Trumpet Tree

COMMON / LOCAL NAME: Cetico

SPANISH: Cetico

TRADITIONAL MEDICINAL USES:

* Root

* Leaves



The cecropia tree is known as a "pioneer species" because it has a great capacity to grow in almost any type of soil. It is one of the fastest-growing trees in the Amazon.

It grows to approximately 20 meters in height, with a trunk diameter of about 50 centimeters. Its scientific name is Cecropia its local name is cetico, and in English it is called the cecropia tree.

The root of this tree is used for medicinal purposes, mainly as a cardiotonic. In addition, its leaves and flowers serve as a food source for several animal species, such as parrots, oropendolas, and also the sloths, which feeds on its leaves. The trunk is widely used in the paper industry.



HEVEA BRASILENSIS

RUBBER TREE / SHIRINGA

SCIENTIFIC NAME:Hevea brasiliensis

ENGLISH NAME: Rubber Tree

COMMON / LOCAL NAME: Caucho / Shiringa / Jebe

SPANISH: Caucho / Shiringa

The rubber tree, locally known as "caucho" or "shiringa," is very well known in the Loreto region. Its scientific name is Hevea brasiliensis. The tree can grow up to 50 meters tall, with a trunk diameter of 20 to 60 centimeters.

Between the late 19th century and the early 20th century, Iquitos experienced a major economic boom due to the trade of natural rubber. The latex was extracted from rubber trees that are abundant in the Peruvian Amazon.

This boom attracted numerous foreign adventurers and entrepreneurs, mainly from Spain, Italy, Portugal, and Brazil, who settled in Iquitos to take advantage of the commercial opportunities.

During this period, the city expanded rapidly. Many buildings and infrastructure projects were constructed, including the Port of Belén, which was used to export rubber. However, the rubber industry also had a dark side: many Indigenous people and workers were subjected to forced labor and inhumane conditions on the rubber estates, known as cauchales or "rubber centers." Decline and legacy.

By the late 1910s, demand for Amazonian rubber began to decline due to competition from rubber plantations in Asia and the development of synthetic rubber. World War II further disrupted the natural rubber trade, and Iquitos' economy entered a severe recession.

Starting in the 1950s, the city's economy began to diversify into other sectors, such as fishing and oil.

Today, Iquitos preserves several buildings and monuments from its rubber era. Notable examples include the Casa de Fierro, a prefabricated metal structure often attributed to Gustave Eiffel, and the General Cemetery of Iquitos, where many historical figures from the rubber era are buried.





ASPIDOESPERMA PARVIFOLIUM

REMOCASPI

SCIENTIFIC NAME: Aspidosperma parvifolium

ENGLISH NAME: Remo Caspi Tree, Paddle Wood

COMMON/LOCAL NAME: Remocaspi, Remo Caspi

Is a large hardwood tree species found in the tropical rainforests of the Amazon Basin, including Brazil, Colombia and Peru. Here it is very common in terra firme forest and we use it a lot for construction.

It has a straight trunk with grayish-brown bark that is slightly cracked. The wood is very hard, heavy and resistant to water.

Its most special feature is its wood. It is so hard and strong that local people use it to make boat paddles, that is why it is called "Remo Caspi", which means "paddle tree" in Quechua.

That's why it is one of the most valuable trees for local communities.

It measures about 20 - 35 meters in height, with a trunk diameter of 50 - 90 cm.

The trunk is cylindrical and without branches in the lower part.

It lives in well-drained soils and can live for more than 100 years, growing slowly but very strong.

It produces small white flowers and fruits with flat winged seeds that fly with the wind.

Because its seeds are light, the wind disperses them far away to grow new trees.

It flowers to attract bees and other insects that help with pollination.

That process is important to keep the forest healthy and full of life.

It is not a dangerous tree for humans. The wood is safe and very useful.

Its main threat is illegal logging, because people cut it a lot for timber. Living deep in the forest and growing hard wood is its best protection, but it also makes it a target.

The tree starts to produce seeds at 15 - 20 years old. Seed production happens once a year.





PSYCHOTRIA ELATA

KISS FLOWER / HOT LIPS

SCIENTIFIC NAME:Psychotria elata

ENGLISH NAME: Kiss Flower / Hot Lips Plant

COMMON / LOCAL NAME: Labios de Novia / Flor de Beso

SPANISH: Labios de novia

USES:

* Ornamental Cultivation

* Gardening

Psychotria elata is commonly known as the "Kiss Flower" or "Hot Lips" because its bracts are shaped like a pair of bright red, painted lips about to kiss.

It grows from sea level up to 2,000 meters above sea level. It is usually found at the base of trees in tropical forests, and on the banks of springs and streams.

It is a tropical plant native to the jungles of Central and South America, including Colombia, Panama, Ecuador, and Costa Rica.

The plant has evolved this striking shape to attract hummingbirds and butterflies, which act as its pollinators. The "lips" are actually two large, red bracts that protect the small, white and orange flowers that grow between them. The bracts only have this shape for a short period during the flowering phase, before they open.

It is not commonly commercialized because it is currently threatened by deforestation in its natural habitat.

Psychotria elata belongs to the Rubiaceae family. It is a small, herbaceous shrub. It requires a tropical climate with high humidity and frequent rain. It typically grows under the forest canopy, in filtered light.





TRIPLARIS AMERICANA

TANGARANA TREE

SCIENTIFIC NAME:Triplaris americana L.

ENGLISH NAME: Tangarana Tree

COMMON / LOCAL NAME: Tangarana / Palo de Hormiga

SPANISH: Tangarana

TRADITIONAL MEDICINAL USES:

* Leaves

* Bark

The tangarana tree can reach a height of 30 meters. Its scientific name is Triplaris americana. It has simple, alternate leaves and cylindrical branches with hollow sections, called domatia, where ants live.

It has a symbiotic relationship with an aggressive ant species commonly called the "tangarana ant" or "fire ant". The ant is Pseudomyrmex viduus, measures about 5 millimeters, and is reddish in color. It lives inside the hollow branches of the tree and defends it from herbivores.

The ant has a very painful sting. In some 2 the morning as a traditional practice to feel more alert before starting their daily activities. This has led to the tree being nicknamed "Amazon coffee" or "jungle coffee."

The leaves are used to relieve severe burns. The bark of the trunk is used to treat malaria, and bark infusions are used against diarrhea. The plant contains tannins, flavonoids, and steroids. Dried, smoked leaves have been reported to be used as a hallucinogen.



FERNS

Ferns belong to the group of spore-producing plants, meaning that they reproduce through spores rather than seeds. They were among the first higher plants to appear on the planet and provide important ecological services, such as oxygen production, water retention and purification, and erosion control on steep slopes.

Additionally, their elegant appearance has earned them a prominent role as ornamental plants in parks and gardens. Some species are also cultivated as complementary foliage in the floral industry.



Fern Reproduction

Ferns are the only vascular plants that do not produce seeds. Of course, they do not have flowers either. So, how do they reproduce?

In general, ferns go through two stages during their life cycle: one reproduces sexually and is called the gametophyte, while the other is called the sporophyte and reproduces asexually through spores.

The plant we call a fern, which we commonly keep at home for the beauty of its fronds, is the sporophyte.

If we carefully examine the underside of the leaves of a sword fern from late winter and throughout spring and summer, we can observe brown structures called sporangia, within which the spores develop.

These sporangia may cover the entire abaxial surface of the leaf (the underside), occur only along the edges

and be protected by the curved margin of the leaf blade, or be grouped into well-defined structures called sori. These sori may or may not be protected by a special structure known as an indusium.

Sporangia can produce one type of spore (in homosporous plants) or two types: megaspores and microspores. In the latter case, the plant is called heterosporous, a characteristic of the family Salviniaceae, and it has megasporangia or microsporangia, depending on the type of spore produced.

When the spores produced by the sporophyte fall in a place with suitable warmth and moisture, they begin to germinate and develop into the gametophyte. This structure is much smaller than the sporophyte and is completely independent of it. It generally has a much shorter lifespan and is known as the prothallus.

The sexual organs are found on the prothallus: the female organ, called the archegonium, and the male organ, called the antheridium. In some species, these organs are found on separate prothalli.

The prothallus has a very short lifespan and survives only until fertilization of the eggs in the archegonia takes place. This process can occur only in the presence of water, which is why ferns are highly sensitive to environmental changes.

The fertilized egg develops into a new sporophyte, which will form its rhizome, underground stem, and fronds.

It is as though, during one part of its life, the plant has one form—the familiar fern we all recognize—and during another part of its life, it takes on a completely different form.





EPIPHYTIC PLANTS

Tropical forests are home to an important group of plants known as epiphytes, which contribute to biodiversity, provide resources for wildlife (such as shelter, food, and breeding sites), and play a role in intercepting moisture and nutrients from the atmosphere.



Epiphytic Plants

The word epiphyte (from the Greek epi, meaning “upon,” and phyton, meaning “plant”) refers to plants that grow on another plant organism, called a phorophyte, which they use solely for support. The roots of the epiphyte attach themselves to the phorophyte but do not obtain any nutrients from it.

Nutrients come from sources such as dust and rain (which collects them from the air, where they are suspended). In addition, epiphytes take advantage of the decomposition of leaves and other dead organisms or fragments of them. They also capture water that runs down the trunk when it rains, carrying minerals and organic matter from the tree canopy.

Some bromeliads belonging to the subfamily Bromelioideae, such as Aechmea, Neoregelia, and Billbergia, modify the arrangement of their leaves, organizing them into a funnel-shaped rosette that allows them to store water and organic matter in the center. For this reason, they are known as tank bromeliads.

Orchids, aroids, and some palm species have developed specialized tissue covering their roots, known as the velamen. This is a type of epidermis made up of layers of dead cells with thickened cell walls. The velamen protects the roots from damage and absorbs water during the rainy season, while during the dry season it provides a

barrier that prevents water loss through transpiration.

Epiphytic plants are the result of evolutionary adaptations that allow certain organisms to survive in diverse ecosystems, such as tropical rainforests and humid tropical forests.

Epiphytism involves approximately 10% of plant species. These include many fern species, as well as plants from several families, such as Orchidaceae, Bromeliaceae, and, to a lesser extent, families such as Piperaceae, Cactaceae, Ericaceae, and Rubiaceae, among others.

Epiphytes are often mistakenly blamed for being parasites. However, we can sometimes observe them growing on dead trees, electrical wires, or rocks. So, is it true that they are parasites? Clearly, these "objects" do not contain sap, water, or nutrients that the plants could obtain, making it impossible for the epiphyte to be a parasite in these situations.

Parasitic Plants

Parasitic plants are those that obtain some or all of the substances they need for growth, including water and nutrients, from another plant called the host.

Some parasitic plants, known as hemiparasites, contain chlorophyll and obtain only part of their nutrients from the host. Another group consists of plants that lack chlorophyll, are usually whitish in color, and obtain all their nutrients from the host. These are called holoparasites.

Holoparasites infect the host plant through specialized roots called haustoria, which penetrate the outer tissue and connect to the vascular tissue. This process can eventually cause the death of the host plant.





BROMELIADS

Bromeliads, belonging to the family Bromeliaceae, comprise approximately 3,086 species worldwide (Luther, 2006). In Mexico, 342 species are known, and the state of Oaxaca ranks first nationally, with 172 bromeliad species.

Bromeliads are used as ornamental plants in Nativity scenes, altars, and patron saint festivities. They are also used as animal fodder, and their fibers and medicinal products are extracted for various purposes.

TYPES OF EPIPHYTIC BROMELIADS

These plants obtain nutrients through their leaves, while their roots serve to anchor them to the branches or trunks on which they grow.

Epiphytic bromeliads can be classified into two types:

- **Tank bromeliads:** They have large, non-pubescent leaves arranged in rosettes. The overlapping leaves form a tank where water and leaf litter accumulate. They require higher moisture levels in the tank and grow best in less sunny locations.

- **Atmospheric bromeliads:** They have thin leaves covered with abundant trichomes (tiny, hair-like structures) that help absorb water and nutrients. They do not tolerate excessive moisture and are typically found in areas more exposed to sunlight.

Bromeliads provide shelter and food for different types of animals. In addition, they contribute nutrients to the soil and help maintain moisture in the forest.

Because of their beautiful foliage and flowers, ease of cultivation, and resistance to many pests, numerous bromeliad species have significant ornamental value.





Tank Bromeliads

These plants are 40 to 50 cm tall and inhabit both humid and dry climates. With the help of their elongated leaves, arranged in a rosette, they can store large quantities of leaf litter, soil, and water for their nourishment.

Atmospheric Bromeliads

These plants are 10 to 15 cm tall and inhabit warm areas. Their leaves have a large number of trichomes, which help them retain moisture and absorb nutrients.

Sexual Reproduction

This process involves the male and female reproductive organs, which are found inside the flowers that make up the plant's inflorescences. Through sexual reproduction, bromeliads can produce and disperse seeds.

Advantages:

- Allows plants to become established over larger areas.
- Provides greater resistance to diseases and pests.
- Enables plants to adapt better to their environment.

Disadvantages:

- Slow growth.
- Low survival rates during early development because animals may feed on the plants or seeds during germination.

Asexual Reproduction

The mother plant produces offshoots, or pups, which can develop into large colonies.

Advantages:

- Rapid growth.
- Each plant can produce between one and five offshoots, depending on the species.

Disadvantages:

- Bromeliads may fall due to the weight of their numerous offshoots.
- They have lower resistance to pests and diseases.

What Happens When Epiphytic Bromeliads Fall?

Epiphytes can fall for several reasons, including becoming detached, the breaking of a branch, or the falling of the tree on which they grow.

When this happens, epiphytic bromeliads may die within less than a year, depending on environmental conditions and the size of the plant.

Therefore, this manual focuses on making use of fallen bromeliads before they die by collecting and replanting them.

HELICONIAS

Heliconias are remarkable plants, not only because of the beauty of their flowers but also because so little is known about them. Anyone traveling to the majestic department of Vaupés cannot help but notice these tall, exotic plants when they are in bloom, as they can be found throughout the department.

These beautiful flowers are native plants that are easy to grow and have become favorites in gardens throughout the municipality of Mitú. They have also become very popular as cut flowers.

Distribution and Habitat

The genus Heliconia (200–250 species, family Heliconiaceae) includes large, rhizomatous, herbaceous perennials distributed primarily throughout the humid forests of the Neotropics, although some species are also found in the Pacific Basin and extend into Southeast Asia (Berry & Kress, 1991; Kress, 1984, 1990).

The greatest diversity of species occurs in montane forests and cloud forests, but the most vigorous growth is found in the lowlands of the Neotropics. There, heliconias are common pioneer plants in successional habitats along roadsides, riverbanks, and gaps created by fallen trees in rainforests.

In large open habitats characterized by high solar radiation (Chazdon & Fetcher, 1984; Chazdon et al., 1996), polyaxial heliconias can exceed 6 meters in height and form dense stands containing 50 or more shoots.

The most robust Heliconia clones exhibit a musoïd growth form, characterized by pseudostems with semi-erect leaves. This growth form closely resembles that of successional species belonging to the related families Musaceae and Strelitziaceae, which occur in the tropics of the Old World (order Zingiberales; Dahlgren et al., 1985).

The genus Heliconia also includes species with a zingiberoid growth form, characterized by horizontal, distichous leaves produced along a firm stem. These species often colonize small forest gaps. Other species are low-growing, nearly stemless herbs that inhabit the understory of mature, climax forests.

Although leaf morphology is relatively uniform throughout the genus, leaf anatomy follows a basic structural pattern (Tomlinson, 1959, 1969). However, interesting variations occur in the tissues involved in photosynthesis, including differences in leaf blade thickness and mesophyll structure in Heliconia (Triplett & Kirchoff, 1991). These characteristics may be related to the light conditions preferred by different species in their native habitats (Rundel et al., 1998).

The genus Heliconia is centered in South America and Central America, with its main center of variation extending along the Andes from Peru to Costa Rica. Four of the five subgenera—Heliconia, Griggsia, Stenochlamys, and Taeniostrobus—are restricted to the Neotropics. The fifth subgenus, Heliconiopsis, occurs in the western Pacific,

including Fiji, Samoa, and the Solomon Islands (Kress, 1990b; Andersson in Kubitzki, 1998, “Heliconiaceae”).

Uses

Most Heliconia species are cultivated as ornamental plants, although the rhizomes or tubers of some species can be eaten roasted or cooked.

One example is Heliconia bihai, which is widely used in the preparation of regional dishes. Its leaves serve as wrappers for tamales and juanes, traditional foods in parts of Latin America.

Another species, Heliconia rostrata, commonly known as patujú, is the national flower of Bolivia.

Heliconias also help protect water sources and play an important role in reforestation efforts.



KINGDOM FUNGI

Fungi are neither plants nor animals; instead, they are classified in a separate kingdom known as Kingdom Fungi. When we think of a fungus, the first thing that usually comes to mind is a mushroom with a cap, such as the common mushroom (Agaricus bisporus). However, this type of morphology is just one of the many forms fungi can take.

The variety of shapes, colors, and sizes found among fungi is immense. They can grow in shelf-like forms, as seen in Trametes versicolor and Pycnoporus sanguineus on standing or fallen tree trunks; in cup-shaped forms, as in Cookeina colensoi and Ascobolus scatigenus; or even in star-shaped forms, as in Geastrum saccatum.

However, the actual body of a fungus, known as the mycelium, is composed of long filaments called hyphae. If we examine the substrate on which fungi grow, such as soil or wood, under a microscope, we will find that it is permeated by hyphae.

What we commonly call a “mushroom” is the fruiting body of the fungus, which is responsible for producing spores used in reproduction. The spore-producing structures are not distributed throughout the entire fruiting body; instead, they are organized on a specific surface called the hymenium. The region where the hymenium is located, together with its supporting structures, is known as the hymenophore.

Other species produce a mass of spores called the gleba, as occurs in gasteroid fungi. In these cases, a distinct hymenophore cannot be identified in mature fruiting bodies.

Differences in the hymenophore are among the main characteristics used to determine the type of fungus being examined.

This article will focus on two major fungal groups: Ascomycota and Basidiomycota. We will also discuss some of the different types found within these groups, based on the morphology of their fruiting bodies.

The goal is to provide guidance for people who encounter a fungus in nature and want to identify it. This is not always easy, as identification may require a magnifying glass or microscope, specific chemical reagents, or even molecular tools.

Nevertheless, knowing how to recognize the major fungal group to which a specimen belongs provides a useful starting point. It helps us determine where to look next, which structures to examine closely, and which reference materials may be useful.





AMPHIBIAN WORLD VILLA HERMOSA FARM

Amphibian Diversity and Conservation

Fundo Villa Hermosa is a natural environment that provides habitat for a variety of amphibians, including frogs, toads, and salamanders. These animals play a fundamental role in maintaining ecological balance in tropical ecosystems. They inhabit a range of microhabitats, including moist soil, leaf litter, ponds, streams, and areas with dense vegetation.

Amphibians are particularly sensitive to environmental changes because their skin is permeable and many species depend on moist environments for reproduction and development. Their presence and diversity can provide valuable information about the condition of local ecosystems.

These animals contribute to ecological balance by controlling insect and other invertebrate populations while also serving as a food source for reptiles, birds, mammals, and other predators. Their life cycles, which often include aquatic or terrestrial larvae and adult stages, connect different components of the ecosystem.

Exploring the amphibian diversity of Fundo Villa Hermosa offers an opportunity to appreciate the richness of tropical biodiversity and understand the importance of conserving natural habitats. Protecting water sources, preserving forest vegetation, maintaining leaf litter, and avoiding pollution are essential measures for ensuring the survival of these fascinating animals.

Amphibian Species Recorded at Fundo Villa Hermosa

The following list presents the amphibian species reported for Fundo Villa Hermosa, organized by family.

1. Aromobatidae

- *Allobates trilineatus*

2. Bufonidae

- *Atelopus harlequin*
- *Rhaebo ceratophrys*
- *Rhinella margaritifera*

3. Dendrobatidae

- *Ameerega hahneli*
- *Ranitomeya amazonica*
- *Ranitomeya reticulata*

4. Hylidae

- *Boana calcarata*
- *Boana lanciformis*

5. Leptodactylidae

- *Adomera andreae*
- *Adenomera hylaedactyla*
- *Leptodactylus pentadactylus*

6. Plethodontidae

- *Bolitoglossa altamazonica*

7. Strabomantidae

- *Oreobates quixensis*
- *Pristimantis altamazonicus*
- *Pristimantis malkini*



Ecological Importance of Amphibians

The amphibians of Fundo Villa Hermosa contribute to the functioning of tropical ecosystems in several ways:

- **Natural pest control:** Many species feed on insects and other small invertebrates, helping regulate their populations.
- **Food-web relationships:** Amphibians are an important food source for numerous predators, supporting the flow of energy through the ecosystem.
- **Environmental indicators:** Because many amphibians are sensitive to changes in temperature, moisture, water quality, and habitat structure, monitoring their populations can help reveal environmental disturbances.
- **Nutrient cycling:** Through feeding, growth, reproduction, and decomposition, amphibians contribute to nutrient movement between aquatic and terrestrial habitats.
- **Biodiversity conservation:** Maintaining suitable habitats for amphibians also helps protect the many other organisms that share their environment.

Habitat and Adaptations

Tropical amphibians occupy a wide range of ecological niches. Some species are commonly found among fallen leaves on the forest floor, while others use vegetation, tree cavities, or temporary pools as shelter and breeding sites.

Species of *Pristimantis*, for example, are known for direct development in many cases, meaning that their eggs hatch into small froglets rather than free-swimming tadpoles. Other frogs depend on water or moist terrestrial environments for larval development. These differences in reproductive strategies allow amphibians to occupy diverse habitats.

Salamanders such as *Bolitoglossa altamazonica* also contribute to the diversity of the region's amphibian fauna. Like frogs and toads, they depend on suitable environmental conditions to survive, making the conservation of humid forest habitats especially important.

Threats and Conservation

Amphibians face several threats, including deforestation, habitat fragmentation, water pollution, climate change, and infectious diseases. Changes in forest structure and moisture levels can affect their feeding, reproduction, and survival.

Conservation efforts at Fundo Villa Hermosa can include protecting streams and wetlands, retaining leaf litter and fallen logs, avoiding the use of harmful chemicals, and minimizing disturbance to breeding areas. Regular monitoring and careful documentation of species can also help improve knowledge of local biodiversity.

When observing amphibians, visitors should avoid handling them unnecessarily, since their permeable skin can be easily damaged by oils, chemicals, and residues on human

hands. Photographs and field notes provide useful ways to document observations without disturbing the animals.

Discovering and Protecting Amphibian Diversity

The amphibians associated with Fundo Villa Hermosa represent an important component of tropical biodiversity. Learning to recognize their species, understanding their ecological roles, and protecting the habitats on which they depend can contribute to long-term conservation.

Every observation can help increase our understanding of these animals and their environment. By preserving the natural conditions that amphibians require, we also help safeguard the ecological processes and biological richness of the tropical forest.

Allobates ***trilineatus*** (Boulenger 1884)

Rana saltarina trilineada / Striped Rocket Frog

Familia: Aromobatidae **Género**: Allobates





Atelopus ***harlequin***

(Coloma, Plewnia, Böning, Boistel, Ellwein, Lötters, Paluh, Roca-Rey Ross and Venegas, 2025)

Harlequin Jambato

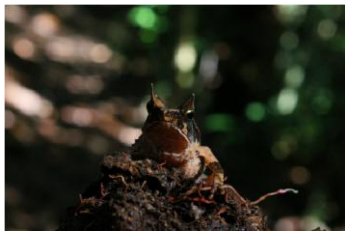
Familia: Bufonidae **Género**: Atelopus





Rhaebo *ceratophrys* (Boulenger 1882)
Sapo cornudo termitero / Termite horned frog

Familia: Bufonidae **Género**: Rhaebo





Rhinella *margaritifera* (Laurenti 1768)
Sapo común sudamericano / South American Common Toad

Familia: Bufonidae **Género**: Rhinella





Ameerega *hahneli* (Boulenger 1884)
Rana venenosa de Yurimaguas / Yurimaguas Poison Dart Frog

Familia: Dendrobatidae **Género**: Ameerega





Ranitomeya *amazónica* (Schulte 1999)

Ranita venenosa amazónica / Amazonian Poison Frog

Familia: Dendrobatidae **Género**: Ranitomeya





Ranitomeya *reticulata* (Boulenger 1884)

Rana venenosa rojiza / Red-backed poison frog

Familia: Dendrobatidae **Género**: Ranitomeya





Boana *fasciata* (Günther 1858)

Rana arbórea de Gunther / Gunther's Banded Treefrog

Familia: Hylidae **Género**: Boana





Boana *lanciformis* (Cope 1871)

Rana lanceolada común / Basin treefrog

Familia: Hylidae **Género**: Boana





Adenomera *andreae* *(Müller 1923)*

Rana terrestre de André / Lowland tropical bullfrog

Familia: Leptodactylidae **Género**: Adenomera





Adenomera *hylaedactyla* (Cope 1868)

Rana terrestre del Napo / Napo Tropical Bullfrog

Familia: Leptodactylidae **Género**: Adenomera





Leptodactylus *pentadactylus* *(Laurenti 1768)*

Hualo, Rana Ternero / Smoky Jungle Frog

Familia: Leptodactylidae **Género**: Adenomera





Bolitoglossa *altamazonica* (Cope 1874)

Salamandra amazónica / Nauta Mushroom-tongue Salamander

Familia: Plethodontidae **Género**: Bolitoglossa





Oreobates *quixensis* (Jiménez de la Espada 1872)

Sapito bocón amazónico / Common Big - headed Frog

Familia: Strabomantidae **Género:** Oreobates





Pristimantis *altamazonicus* (Barbour y Dunn 1921)
Cutín amazónico / Amazon Robber Frog

Familia: Strabomantidae **Género**: Pristimantis





Pristimantis *malkini* (Lynch 1980)
Cutín del río Ampiyacu / Ampiyacu Robber Frog

Familia: Strabomantidae **Género**: Pristimantis







BIRD WORLD VILLA HERMOSA FARM

BIRD DIVERSITY ALONG THE ECOLOGICAL INTERPRETATION TRAIL OF FUNDO VILLA HERMOSA

Bird Diversity and Ecotourism Potential

The study entitled *"Bird Diversity Along the Ecological Interpretation Trail of the Joaquín Reátegui Medina Public Higher Technological Education Institute – Nauta, 2020"* was presented by students Juan Walter Lomas Pacaya and Hedilberto Amado Romayna Celis as part of the requirements for obtaining their professional technical qualifications.

The researchers concluded that, based on fieldwork and consultations with bibliographic sources, including birdwatching guides, the ecological interpretation trail of the Joaquín Reátegui Medina Public Higher Technological Education Institute (IESTP "JRM") is home to **160 bird species**. This diversity offers considerable potential for the development of birdwatching tourism.

Among the most notable birds mentioned in the study are the Yellow-rumped Cacique, Brown-tailed Parakeet, Blue-crowned Manakin, and Smooth-billed Ani. These species contribute to the area's natural appeal and provide opportunities for birdwatching, environmental education, and wildlife interpretation.

Importance of Accessibility and Tourism Development

Improving access to the ecological interpretation trail is essential to strengthening its potential as a destination for ecotourism and environmental education. The proposed construction of two bridges would facilitate access to the site, allowing visitors, students, researchers, and tourists to explore the flora and fauna of the Amazon more easily, particularly the bird diversity of the Loreto region.

Fundo Villa Hermosa, where the ecological interpretation trail is located, is an important setting for practical field training for students enrolled in the Official Tourism Guiding program at IESTP "JRM." The trail provides opportunities to develop skills in wildlife observation, environmental interpretation, and guided tourism.

Ecological and Educational Value

Birds play essential roles in tropical ecosystems, including seed dispersal, pollination, insect population control, and the maintenance of ecological balance. Their diversity also makes them valuable indicators for environmental education and nature-based tourism.

The documented richness of bird species along the trail highlights the importance of conserving the area's natural habitats and promoting responsible tourism practices. Improving accessibility while protecting the forest and its wildlife could help strengthen local educational activities and create opportunities for sustainable tourism development.

In conclusion, the ecological interpretation trail of Fundo Villa Hermosa represents a valuable resource for biodiversity conservation, environmental learning, and birdwatching tourism in Loreto. Improved access, appropriate interpretation infrastructure, and responsible management could help visitors appreciate the region's natural heritage while supporting the practical education of future tourism professionals.

Source: Lomas Pacaya, Juan Walter, and Romayna Celis, Hedilberto Amado (2020), study on bird diversity along the ecological interpretation trail of IESTP "Joaquín Reátegui Medina," Nauta, Loreto, Peru. The information and species count are presented as reported in the supplied text.

2026

WILDLIFE CONSERVATION

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VILLA HERMOSA FARM



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